

TCS3351 Cryptography and Data Security

Tutorial 5 Questions

Modern Symmetric Ciphers-II (Advanced Encryption Standard (AES))

1. Answer the following questions with respect to Advanced Encryption Standard (AES).
 - a. List the block size, key size, number of rounds, number of round keys, and number of transformations for the three AES versions.
 - b. Define a state in AES. How many states are there in each version of AES?
 - c. If a and b are two bytes, show that
$$\text{SubBytes}(a \oplus b) \neq \text{SubBytes}(a) \oplus \text{SubBytes}(b)$$
Use $a = (35)_{16}$ and $b = (46)_{16}$ as an example.

Answer:

- a) Block size: 128-bit

AES	Key Size	No. of Rounds	No. of Round Keys	No. of Transformation
128	128	10	11	40
192	192	12	13	48
256	256	14	15	56

; round0 original key is used.

- b) A state in AES refers to a data block.

- i) AES-128: 41 states
- ii) AES-192: 49 states
- iii) AES-256: 57 states

- c) $a = (35)_{16} = (00110101)_2$; $\text{SubBytes}(a) = (96)_{16} = (10010110)_2$
 $b = (46)_{16} = (01000110)_2$; $\text{SubBytes}(b) = (5A)_{16} = (01011010)_2$

$$a \oplus b = (01110011)_2 = (73)_{16};$$
$$\text{SubBytes}(a \oplus b) = \text{SubBytes}(73) = (8F)_{16}$$

$$\text{SubBytes}(a) \oplus \text{SubBytes}(b) = (11001100)_2 = (CC)_{16}.$$

Hence, $\text{SubBytes}(a \oplus b) \neq \text{SubBytes}(a) \oplus \text{SubBytes}(b)$.

2. With respect to Advance Encryption Standard, answer the following:

Given the plaintext

(00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F)_H

and the key

(01 01 01 01 01 01 01 01 01 01 01 01 01 01 01 01)_H

- a. Show the original contents of State, displayed as a 4×4 matrix.
- b. Show the value of State after the initial AddRoundKey.
- c. Show the value of State after SubBytes.
- d. Show the value of State after ShiftRows.
- e. Show the value of State after MixColumns.

Answer:

a. The original state

00	04	08	0C
01	05	09	0D
02	06	0A	0E
03	07	0B	0F

b. The output state after initial AddRoundKey:

00	04	08	0C
01	05	09	0D
02	06	0A	0E
03	07	0B	0F

 $\oplus$

01	01	01	01
01	01	01	01
01	01	01	01
01	01	01	01

 $=$

01	05	09	0D
00	04	08	0C
03	07	0B	0F
02	06	0A	0E

c. The output state after SubBytes:

01	05	09	0D
00	04	08	0C
03	07	0B	0F
02	06	0A	0E

 $\rightarrow$

7C	6B	01	D7
63	F2	30	FE
7B	C5	2B	76
77	6F	67	AB

d. The output state after ShiftRows:

1st row: no shift, 2nd row shift 1 byte left, 3rd row, shift 2 byte left, 4th row : shift 4 byte left.

7C	6B	01	D7
F2	30	FE	63
2B	76	7B	C5
AB	77	6F	67

e. The output state after MixColumns:

02	03	01	01
01	02	03	01
01	01	02	03
03	01	01	02

 $\times$

7C	6B	01	D7
F2	30	FE	63
2B	76	7B	C5
AB	77	6F	67

 $=$

75	87	0F	B2
55	E6	04	22
7E 3E	2E	B8	8C
10	15	58	0A

Row	0:	75	87	0F	B2
Row	1:	55	E6	04	22
Row	2:	3E	2E	B8	8C
Row 3:		10 15 58 0A			

(02 * 7c) XOR (03 * F2) xor (01 * 2B) xor (01 * AB)

02 = 0000 0010 = x

7c = 0111 1100 = $x^6 + x^5 + x^4 + x^3 + x^2$

(02 * 7c) = $(x^6 + x^5 + x^4 + x^3 + x^2)x = x^7 + x^6 + x^5 + x^4 + x^3 = 1111 1000$. (F8)

03 = 0000 0011 = $x + 1$

F2 = 1111 0010 = $x^7 + x^6 + x^5 + x^4 + x$

(03 * F2) = $(x^7 + x^6 + x^5 + x^4 + x)(x + 1)$

= $(x^8 + \cancel{x^7} + \cancel{x^6} + \cancel{x^5} + x^2) + (\cancel{x^7} + \cancel{x^6} + \cancel{x^5} + x^4 + x)$; + is xor

= $x^8 + x^4 + x^2 + x$; Note that : $x^8 = x^4 + x^3 + x + 1$

= $\cancel{x^4} + x^3 + \cancel{x} + 1 + \cancel{x^4} + x^2 + \cancel{x}$

= $x^3 + x^2 + 1 = 0000 1101$

2B = 0010 1011

AB = 1010 1011

02x7c	1111	1000		
03 x f2	0000	1101		
2B	0010	1011		
AB	1010	1011		
XOR	0111	0101		
	7	5		

Inverse Mix Column.

<u>0e</u>	<u>0b</u>	<u>0d</u>	<u>09</u>		<u>75</u>	<u>47</u> <u>87</u>	<u>0F</u>	<u>B2</u> <u>A9</u>		<u>7C</u>	6B	01	D7
				×	<u>55</u>	<u>E6</u> <u>FD</u>	<u>04</u> <u>1F</u>	<u>22</u> <u>39</u>	=	F2	30	FE	63
					<u>7E</u> <u>3E</u>	2E	B8	<u>8C</u> <u>97</u>		2B	76	7B	C5
					<u>10</u>	<u>15</u> <u>0e</u>	<u>58</u> <u>43</u>	<u>0A</u> <u>11</u>		AB	77	6F	67

0e = 0000 1110 = $x^3 + x^2 + x$

75 = 0111 0101 = $x^6 + x^5 + x^4 + x^2 + 1$

(0e * 75) = $(x^3 + x^2 + x)(x^6 + x^5 + x^4 + x^2 + 1)$

= $x^3(x^6 + x^5 + x^4 + x^2 + 1) + x^2(x^6 + x^5 + x^4 + x^2 + 1) + x(x^6 + x^5 + x^4 + x^2 + 1)$

= $x(x^8 + \cancel{x^7} + \cancel{x^6} + \cancel{x^4} + \cancel{x^2}) + x(\cancel{x^7} + \cancel{x^6} + \cancel{x^5} + x^3 + x) + x(x^6 + \cancel{x^5} + \cancel{x^4} + \cancel{x^2} + 1)$

= $x(x^8) + x(x^6 + x^3 + x + 1)$

= $x(x^4 + x^3 + x + 1) + x(x^6 + x^3 + x + 1)$

= $x^5 + \cancel{x^4} + \cancel{x^2} + \cancel{x} + x^7 + \cancel{x^4} + \cancel{x^2} + \cancel{x}$

= $x^7 + x^5$

= 1010 0000 = A0

(0e * 75) = A0 = 1010 0000

(0b * 55) = 61 = 0110 0001

(0d * 3e) = 2D = 0010 1101

$$(09 * 10) = 90 = 1001\ 0000$$

$$0111\ 1100 = 7C$$

There is a mistake in the previous answer (matrix), that why the inverse Mix-Column calculation cannot be done.

Note:

(i) SubByte Transformation Table

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
B	E7	CB	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

(ii) Constant matrices used by MixColumns and InvMixColumns

$$\begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} \xleftrightarrow{\text{Inverse}} \begin{bmatrix} 0E & 0B & 0D & 09 \\ 09 & 0E & 0B & 0D \\ 0D & 09 & 0E & 0B \\ 0B & 0D & 09 & 0E \end{bmatrix}$$

C
 C^{-1}

3. With reference to Advanced Encryption Standard (AES-128) key expansion algorithm, identify the first 3 subround keys. Given Key:

$(0F\ 15\ 71\ C9\ 47\ D9\ E8\ 59\ 0C\ B7\ AD\ D6\ AF\ 7F\ 67\ 98)_H$

Answer:

$w_0 = (0F\ 15\ 71\ C9); w_1 = (47\ D9\ E8\ 59);$

$w_2 = (0C\ B7\ AD\ D6); w_3 = (AF\ 7F\ 67\ 98)$

Round 1:

RotWord: Circular byte shift, $w_3 = (7F\ 67\ 98\ AF)$

SubWord: ByteSubstitution = $(D2\ 85\ 46\ 79)$

Adding Round Constant: $(D2\ 85\ 46\ 79) \oplus (01\ 00\ 00\ 00) = (D3\ 85\ 46\ 79)$

Hence, $t_4 = (D3\ 85\ 46\ 79)$

$w_4 = w_0 \oplus t_4 = (0F\ 15\ 71\ C9) \oplus (D3\ 85\ 46\ 79) = (DC\ 90\ 37\ B0)$

$w_5 = w_4 \oplus w_1 = (DC\ 90\ 37\ B0) \oplus (47\ D9\ E8\ 59) = (9B\ 49\ DF\ E9)$

$w_6 = w_5 \oplus w_2 = (9B\ 49\ DF\ E9) \oplus (0C\ B7\ AD\ D6) = (97\ FE\ 72\ 3F)$

$w_7 = w_6 \oplus w_3 = (97\ FE\ 72\ 3F) \oplus (AF\ 7F\ 67\ 98) = (38\ 81\ 15\ A7)$

First round key:

$(DC\ 90\ 37\ B0\ 9B\ 49\ DF\ E9\ 97\ FE\ 72\ 3F\ 38\ 81\ 15\ A7)$

Round 2:

RotWord: Circular byte shift, $w_7 := (81\ 15\ A7\ 38)$

SubWord: ByteSubstitution: $(0C\ 59\ 5C\ 07)$

Adding Round Constant: $(0C\ 59\ 5C\ 07) \oplus (02\ 00\ 00\ 00) = (0E\ 59\ 5C\ 07)$

Hence, $t_8 = (0E\ 59\ 5C\ 07)$

$w_8 = w_4 \oplus t_8 = (DC\ 90\ 37\ B0) \oplus (0E\ 59\ 5C\ 07) = (D2\ C9\ 6B\ B7)$

$w_9 = w_8 \oplus w_5 = (D2\ C9\ 6B\ B7) \oplus (9B\ 49\ DF\ E9) = (49\ 80\ B4\ 5E)$

$w_{10} = w_9 \oplus w_6 = (49\ 80\ B4\ 5E) \oplus (97\ FE\ 72\ 3F) = (DE\ 7E\ C6\ 61)$

$w_{11} = w_{10} \oplus w_7 = (DE\ 7E\ C6\ 61) \oplus (38\ 81\ 15\ A7) = (E6\ FF\ D3\ C6)$

Second round key:

$(D2\ C9\ 6B\ B7\ 49\ 80\ B4\ 5E\ DE\ 7E\ C6\ 61\ E6\ FF\ D3\ C6)$

Round 3:

RotWord: Circular byte shift, $w_{11} = (FF\ D3\ C6\ E6)$

SubWord: ByteSubstitution = $(16\ 66\ B4\ 8E)$

Adding Round Constant: $(16\ 66\ B4\ 8E) \oplus (04\ 00\ 00\ 00) = (12\ 66\ B4\ 8E)$

Hence, $t_{12} = (12\ 66\ B4\ 8E)$

$w_{12} = w_8 \oplus t_{12} = (D2\ C9\ 6B\ B7) \oplus (12\ 66\ B4\ 8E) = (C0\ AF\ DF\ 39)$

$w_{13} = w_{12} \oplus w_9 = (C0\ AF\ DF\ 39) \oplus (49\ 80\ B4\ 5E) = (89\ 2F\ 6B\ 67)$

$w_{14} = w_{13} \oplus w_{10} = (89\ 2F\ 6B\ 67) \oplus (DE\ 7E\ C6\ 61) = (57\ 51\ 15\ 06)$

$$w_{15} = w_{14} \oplus w_{11} = (57\ 51\ 15\ 06) \oplus (E6\ FF\ D3\ C6) = (B1\ AE\ C6\ 60)$$

Third round key:

(C0 AF DF 39 89 2F 6B 67 57 51 15 06 B1 AE C6 60)

References:

1. Behrouz A. Forouzan, "Cryptography and Network Security", First Edition, McGraw-Hill, 2008
2. William Stallings, "Cryptography and Network Security – Principles and Practice", Seventh Edition, Prentice Hall, 2016.

02	03	01	01
01	02	03	01
01	01	02	03
03	01	01	02

×

E1	01	01	01
F2	01	01	01
03	01	01	01
04	01	01	01

=

Y	X	X	X
X	X	X	X
X	X	X	X
X	X	X	X